



North Eastern Mindanao State University



Introduction to Data Mining

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CS327 – Data Mining

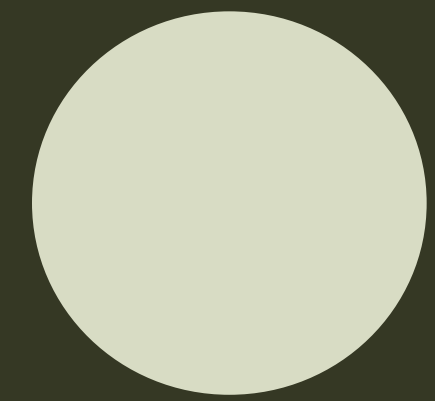


What's inside

- ❑ What is data mining?
- ❑ Data mining on different data
- ❑ Data mining functionalities

Course Intended Learning Outcomes

- ☐ At the end of the session, the students are expected to
 - ☐ Understand the concept of DM, DBMS and ML.
 - ☐ Classify predictive and descriptive ML tasks



What is data mining?

Why Data Mining?

- ❑ *We are living in the age of data*
- ❑ The Explosive Growth of Data: from terabytes to petabytes
- ❑ Data collection and data availability
 - ❑ Automated data collection tools, database systems, Web, computerized society
- ❑ **Major sources of abundant data:**
 - ❑ Business: Web, e-commerce, transactions, stocks, feedback, ..
 - ❑ Science: Remote sensing, bioinformatics, simulations, ..
 - ❑ Society and everyone: news, digital cameras, YouTube, ...
 - ❑ Web searches, social media, telecommunication, medical, ..
- ❑ *We are drowning in data, but starving for knowledge!*
- ❑ *Necessity is the mother of invention*

Evolution of Sciences

- ❑ Before 1600: **Empirical science**

- ❑ 1600-1950s: **Theoretical science**

- ❑ Each discipline has grown a theoretical component.

- ❑ Theoretical models often motivate experiments and generalize our understanding.

- ❑ 1950s-1990s: **Computational science**

- ❑ Over the last 50 years, most disciplines have grown a third, computational branch

- ❑ Computational Science traditionally meant simulation.

- ❑ It grew out of our inability to find closed-form solutions for complex mathematical model

- ❑ 1990-now: **Data science**

- ❑ The flood of data from new scientific instruments and simulations

- ❑ The ability to economically store and manage petabytes of data online

- ❑ The Internet and computing grid that makes all these archives universally accessible

- ❑ Scientific information management, acquisition, organization, query, and visualization tasks scale almost linearly with data volumes. Data mining is a major new challenge!

Evolution of Database Technology

❑ 1960s:

❑ Data collection, database creation, IMS and network DBMS

❑ 1970s:

❑ Relational data model, relational DBMS implementation

❑ 1980s:

❑ RDBMS, advanced data models (extended-relational, object oriented, deductive, etc.)

❑ Application-oriented DBMS (spatial, scientific, engineering)

❑ 1990s:

❑ Data mining, data warehousing, multimedia databases, and Web databases

❑ 2000s:

❑ Stream data management and mining

❑ Data mining and its applications

❑ Web technology (XML, data integration) and global information systems

What is Data Mining?

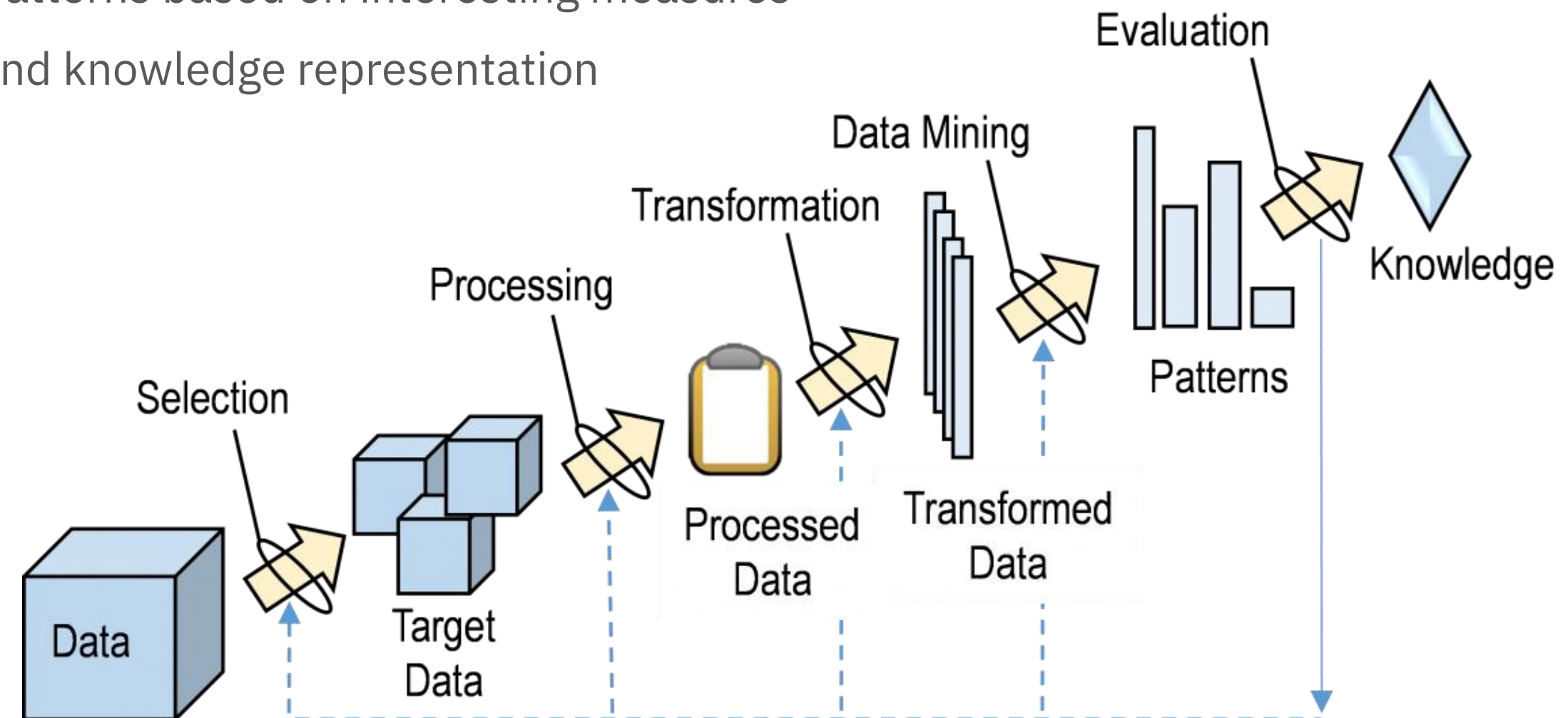
- ❑ Data Mining is the process of discovering interesting, non-trivial, implicit, previously unknown, and potentially useful patterns or knowledge from large volumes of data.
- ❑ Data mining is the core step of the Knowledge Discovery in Databases (KDD) process, involving the extraction of patterns from data.

What is Data Mining?

- ❑ Data mining (knowledge discovery from data)
 - ❑ Extraction of interesting *non-trivial, implicit, previously unknown and potentially useful* patterns or knowledge from huge amount of data
 - ❑ Data mining: a misnomer?
- ❑ Alternative names
 - ❑ **Knowledge** discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- ❑ Watch out: Is everything “data mining”?
 - ❑ Simple search and query processing
 - ❑ (Deductive) Expert systems

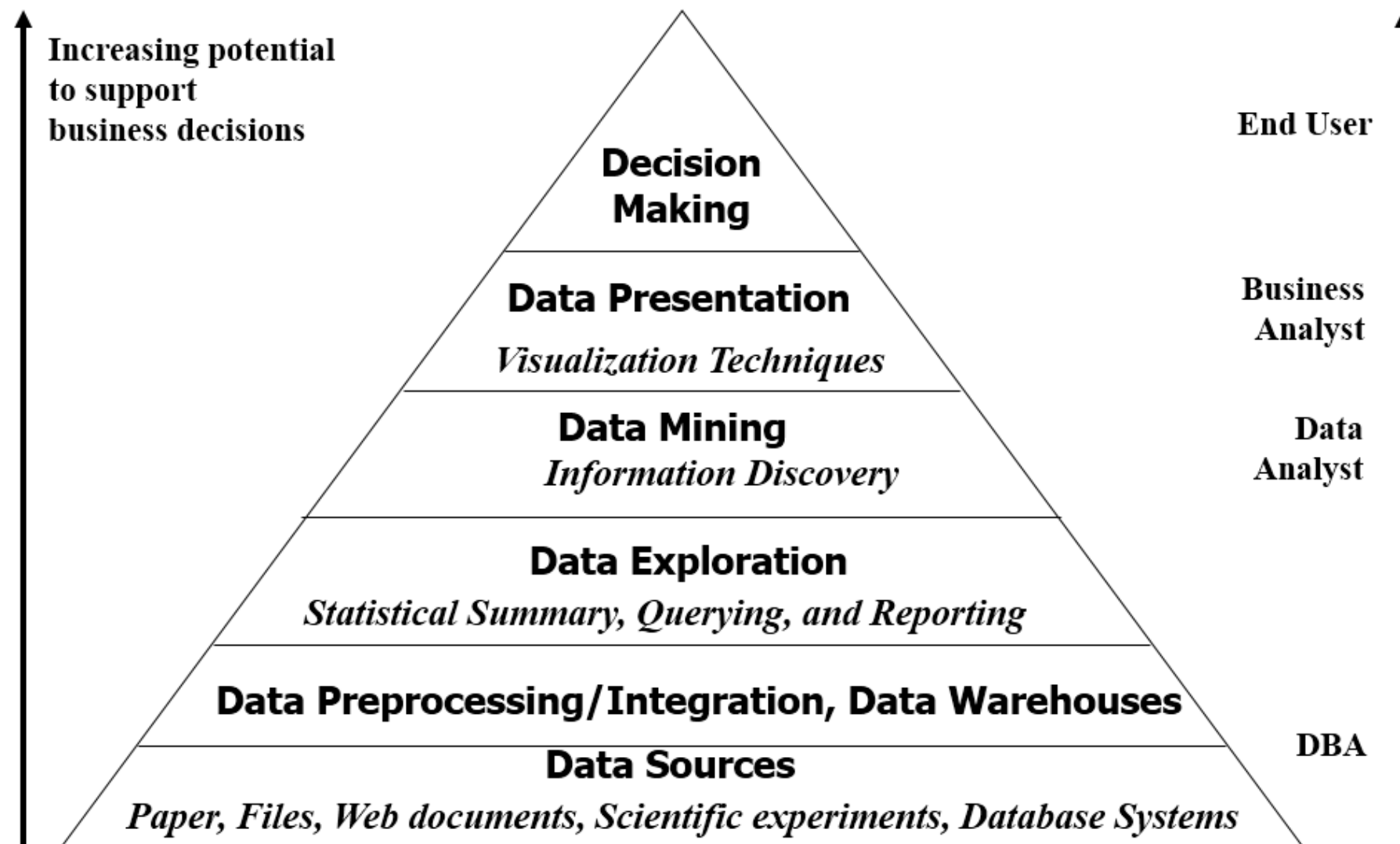
Knowledge Discovery Process

- ❑ Data mining is one stage in the overall KDD process.
- ❑ **Data cleaning:** Remove noise and inconsistent data
- ❑ **Data Integration:** Multiple data sources may be combined
- ❑ **Data Selection:** Relevant data are selected
- ❑ **Data Transformation:** Data are transformed in to form appropriate to mining
- ❑ **Data Mining:** Intelligent methods are applied to extract patterns
- ❑ **Pattern Evaluation:** Identify interesting patterns based on interesting measures
- ❑ **Knowledge Presentation:** Visualization and knowledge representation



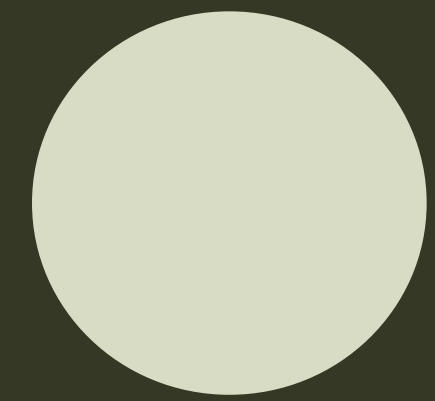
Why Data Mining?

- ❑ Rapid growth of digital data
- ❑ Need for automated analysis
- ❑ Traditional database queries are insufficient
- ❑ Supports decision making in business and science



Data Mining vs Related Concepts

Aspect	Data Mining	Database Systems	Machine Learning	Statistics
Goal	Knowledge discovery	Data storage & retrieval	Model learning	Data analysis
Data Size	Very large	Large	Medium	Small to medium
Focus	Hidden patterns	Queries	Prediction	Inference



Data mining on different data

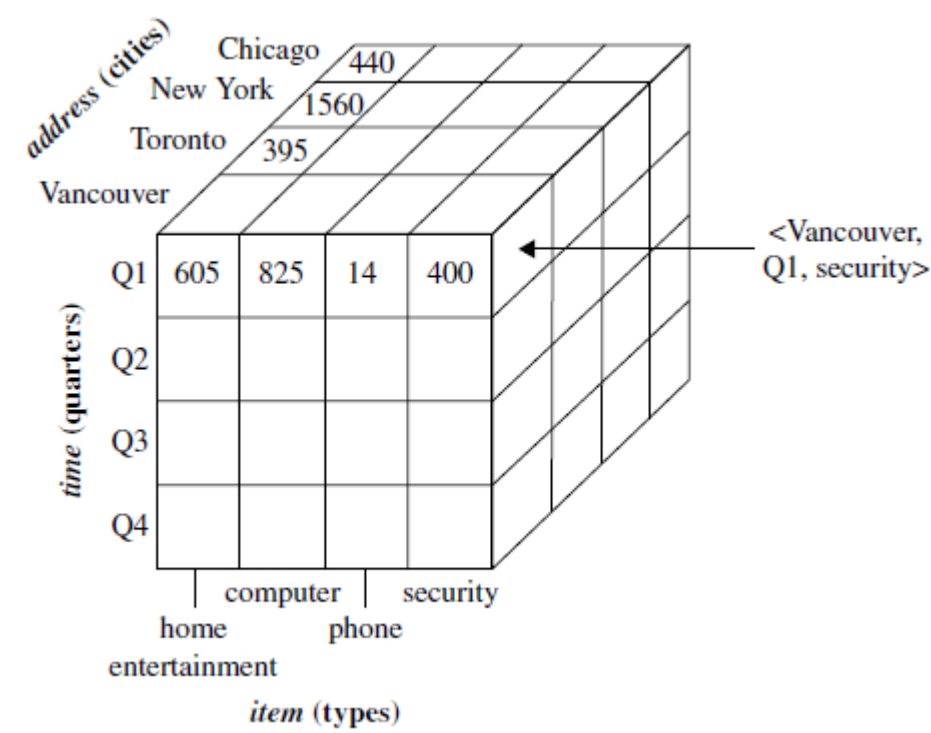
Relational Databases

- ❑ Collection of interrelated data is known as **Database**
- ❑ **Relational database** is collection of tables.
 - ❑ Each table is represented by **tuples** and **attributes**.
 - ❑ Each tuple is uniquely identified by a unique *key*.
 - ❑ Semantic model like **E-R Diagram** represents database as a set of entities and their relationships.
- ❑ **customer** (cust ID, name, address, age, occupation, annual income, credit information, category, . . .)
- ❑ **item** (item ID, brand, category, type, price, place made, supplier, cost, . . .)
- ❑ **employee** (empl ID, name, category, group, salary, commission, . . .)
- ❑ **branch** (branch ID, name, address, . . .)
- ❑ **purchases** (trans ID, cust ID, empl ID, date, time, method paid, amount)
- ❑ **ItemsSold** (trans ID, item ID, qty)
- ❑ **worksAt** (empl ID, branch ID)

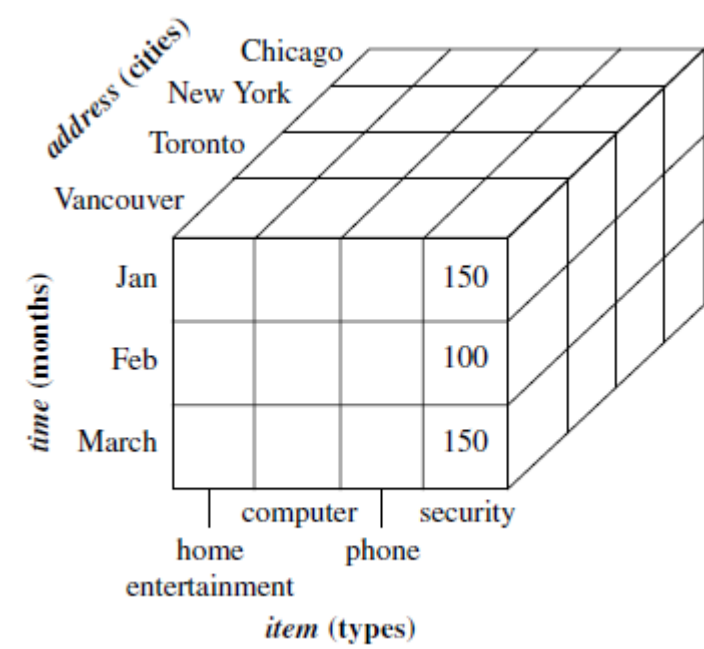
Data Warehouse

- ❑ A data warehouse is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.
- ❑ Rather than storing each transaction, data warehouse may store summary.
- ❑ Data warehouse is typically modeled using multi dimensional data structure called **data cube**
- ❑ In data cube, ***dimension*** corresponds to an attribute, and each ***cell*** stores the value of some aggregate measure such as *count*

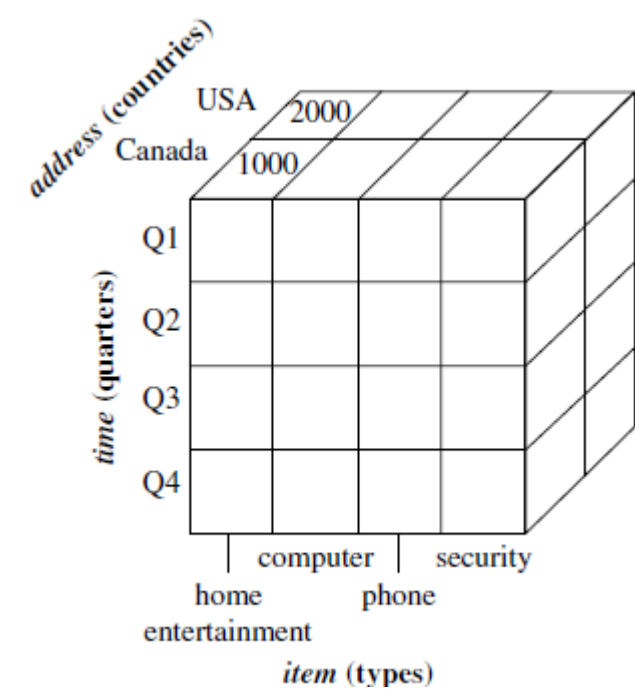
Data Warehouses: OLAP



Data Cube



Drill Down



Roll Up

Transactional Data

- ❑ Record in a transactional database captures a transaction, such as a customer's purchase, a flight booking, or a user's clicks on a web page.
- ❑ Each transaction is a set of items
- ❑ It may contain additional table for other information like sales person, branch etc.
- ❑ Enables **Market basket analysis**: Which items sold well together?
- ❑ **Common Technique:**
 - ❑ Association Rule Mining
 - ❑ Apriori Algorithm

TransactionId	ListofItemsId
T101	I1, I2, I4, I7
T102	I2, I3, I6
...	

Data Mining on Spatial Data

- ❑ Data related to geographic locations
- ❑ Includes maps, satellite images
- ❑ **Applications:**
 - ❑ Urban planning
 - ❑ Climate analysis
 - ❑ GIS systems

Time-Series and Sequence Data

- ❑ Data collected over time
- ❑ Order of events is important
- ❑ **Applications:**
 - ❑ Stock prices
 - ❑ Sensor data
 - ❑ Medical monitoring
- ❑ **Techniques:**
 - ❑ Trend analysis
 - ❑ Sequential pattern mining

Text Data

- ❑ Unstructured data
- ❑ Large volume of documents
- ❑ **Applications:**
 - ❑ Document classification
 - ❑ Sentiment analysis
 - ❑ Information retrieval
- ❑ **Example:**
 - ❑ Classifying research papers into topics

Multimedia Data

- ❑ Images, audio, video
- ❑ Complex and high-dimensional
- ❑ **Applications:**
 - ❑ Image recognition
 - ❑ Video surveillance

Web Data

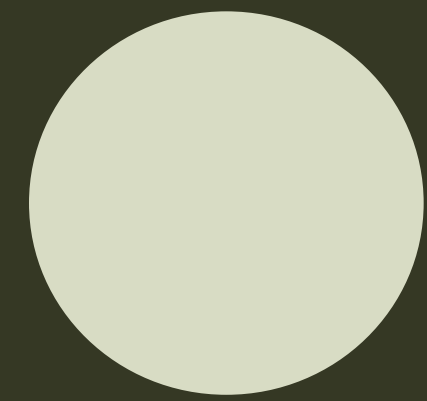
❑ Web pages, hyperlinks, usage logs

❑ **Types:**

❑ Web content mining

❑ Web structure mining

❑ Web usage mining



Data Mining Functionalities

What Kind of Patterns Can be Mined?

- ❑ **Descriptive:** Characterize property of data in target data set
- ❑ **Predictive:** Perform induction on the current data in order to make predictions.
- ❑ Data Mining Functionalities:
 - ❑ Characterization and Discrimination
 - ❑ Mining Frequent Patterns, Associations, and Correlations
 - ❑ Classification
 - ❑ Regression
 - ❑ Cluster Analysis
 - ❑ Outlier Analysis

Characterization and Discrimination

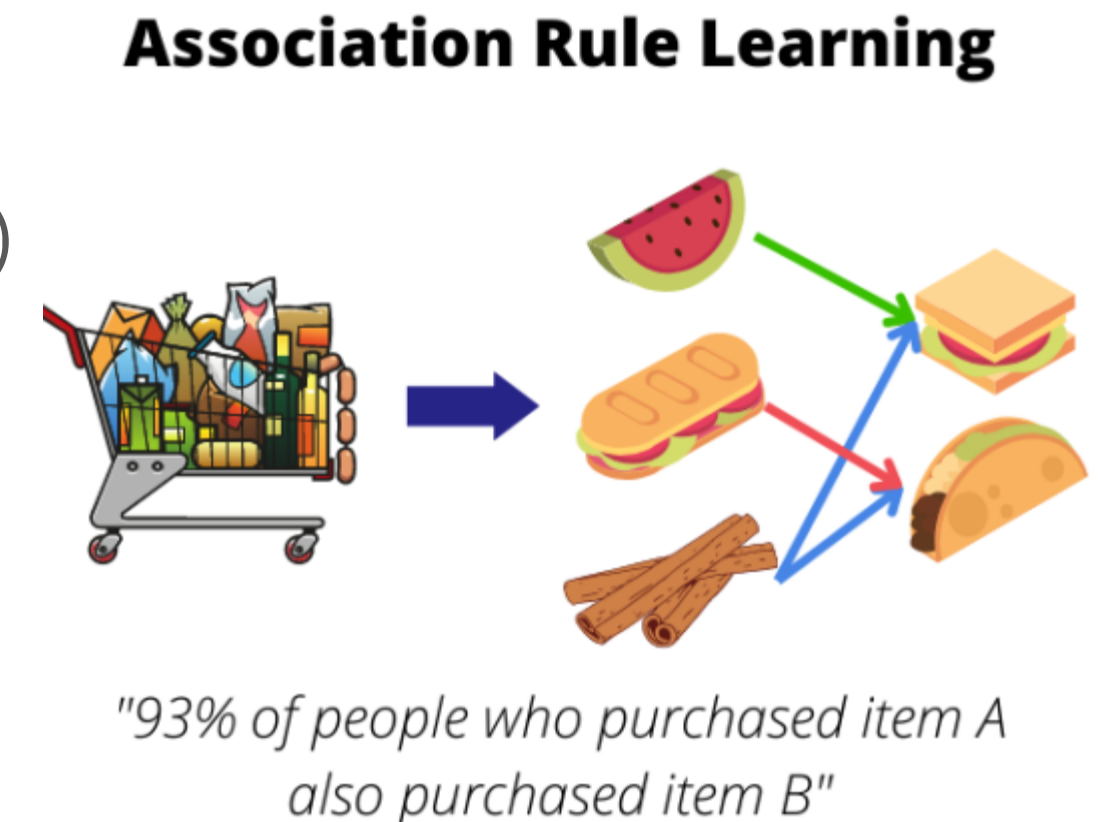
- ❑ **Data characterization** is a summarization of the general characteristics or features of a target class of data.
 - ❑ Summarize the characteristics of customers who *spend more than \$5000 a year* at AllElectronics.
 - ❑ The result is a general profile of these customers, such as that they are 40 to 50 years old, employed, and have excellent credit ratings.
- ❑ **Data discrimination** is a comparison of the general features of the target class
 - ❑ Compare the general features of software products with sales that increased by 10% last year against those with sales that decreased by at least 30% during the same period.

Mining FPs, Association, and Correlation

- ❑ Frequent patterns are patterns that occur frequently in data.
- ❑ **Frequent itemset** typically refers to a set of items that often appear together in a transactional data set—for example, milk and bread
- ❑ A frequently occurring subsequence, such as the pattern that customers, tend to purchase first a laptop, followed by a digital camera, and then a memory card, is a **sequential pattern**
- ❑ **Includes:**
 - ❑ Frequent itemsets
 - ❑ Association rules
 - ❑ Correlations

Association Analysis

- ❑ Single Dimension Association Rule:
 - ❑ $\text{buys}(X, \text{computer}) \rightarrow \text{buys}(X, \text{software})$ [support = 1%, confidence = 50%]
- ❑ A **confidence**, or **certainty**, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy software as well.
- ❑ 1% **support** means that 1% of all the transactions under analysis show that computer and software are purchased together.
- ❑ Multidimensional Association Rule:
 - ❑ $\text{age}(X, 20\ldots 29) \text{ and } \text{income}(X, 40\text{K}\ldots 49\text{K}) \rightarrow \text{buys}(X, \text{laptop})$
 - ❑ [support = 2%, confidence = 60%]



Classification

❓ **Classification** is the process of finding a *model* or *function* that describes and distinguishes data classes.

❓ Construct models (functions) based on some training examples

❓ Assigns data objects to predefined classes

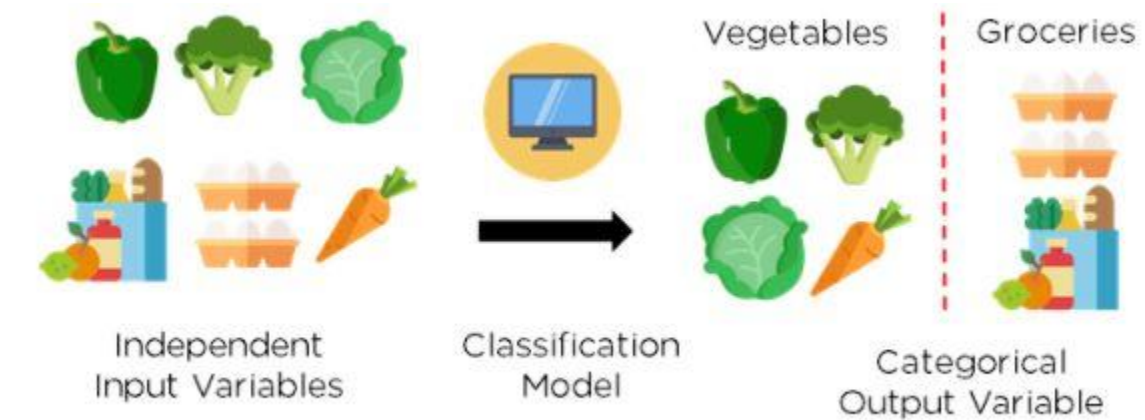
❓ Supervised learning

❓ **Typical methods:**

❓ Decision trees, naïve Bayesian classification, SVM, ANN, rule-based classification, pattern-based classification, LR, etc

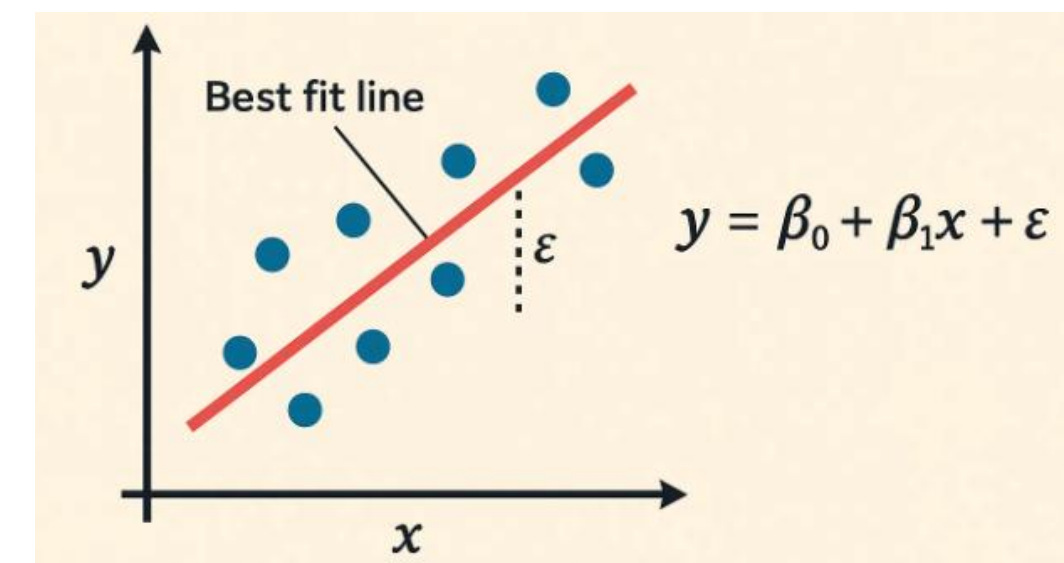
❓ **Applications:**

❓ Credit card fraud detection, Face recognition, classifying flowers, diseases, web-pages, etc



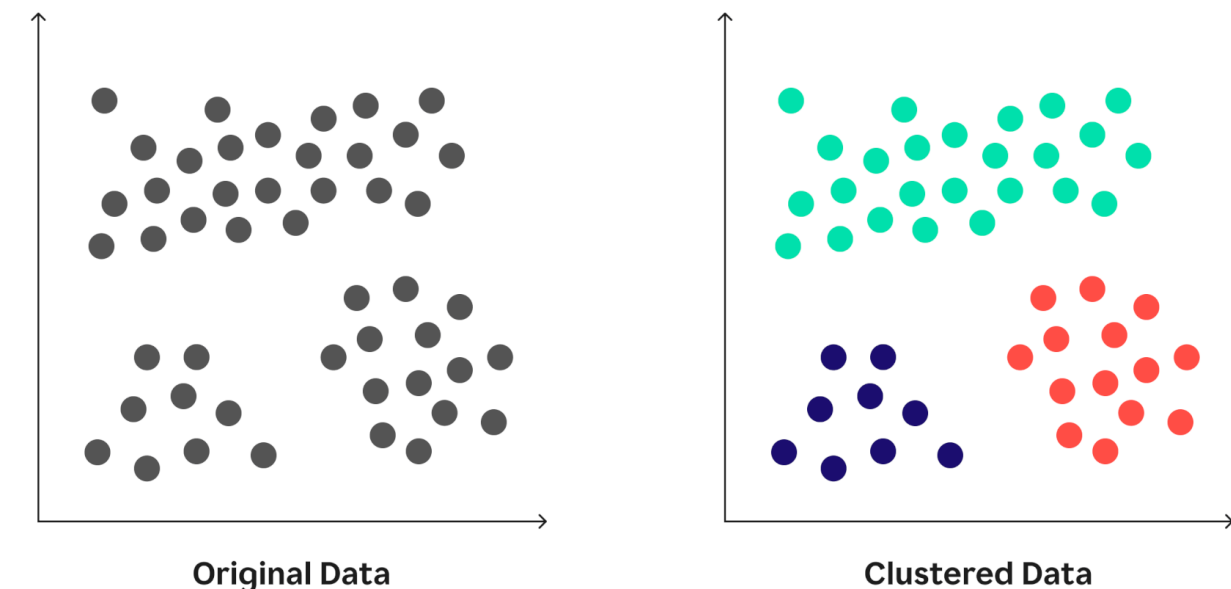
Regression

- ❑ **Regression analysis** is a statistical methodology that is most often used for numeric prediction.
- ❑ Classification predicts categorical (discrete, unordered) labels, regression models predict continuous-valued functions.
- ❑ Classification: Happy, Sad, Fear, Anger
- ❑ Regression: Predict rain in August
- ❑ Classification & regression may need to be preceded by **relevance analysis**, which attempts to identify attributes that are significantly relevant



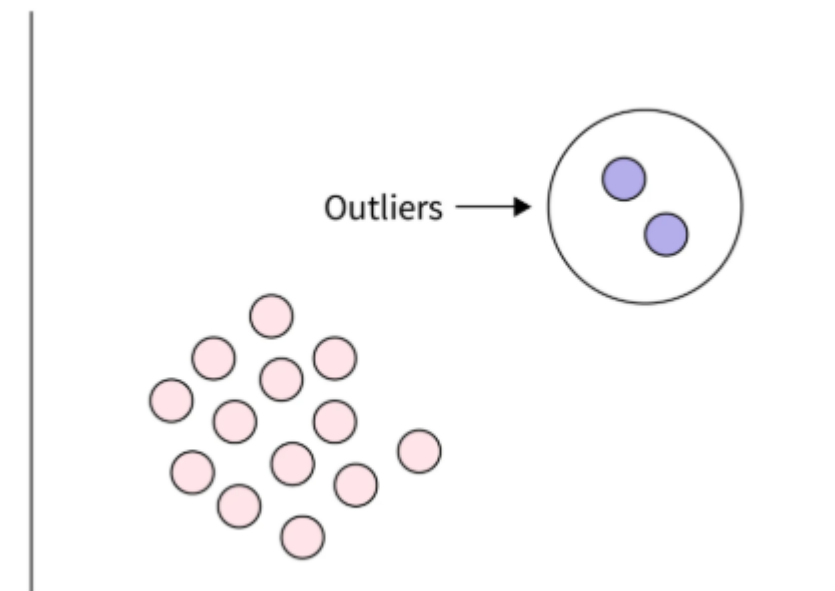
Clustering

- ❑ Unsupervised learning (i.e., Class label is unknown)
- ❑ **Clustering** analyzes data objects without consulting class labels.
- ❑ Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- ❑ **Principle:** Maximizing intra-class similarity & minimizing interclass similarity
- ❑ Clustering can also facilitate **taxonomy formation**.
- ❑ **Techniques:**
 - ❑ k-Means
 - ❑ Hierarchical clustering
- ❑ **Applications:**
 - ❑ Customer segmentation



Outlier Analysis

- ❑ **Outlier / anomaly mining:** A data object that does not comply with the general behavior of the data
- ❑ Outliers may be detected using statistical tests that assume a distribution of the data, or using distance measures where objects that are remote from any other cluster are considered outliers
- ❑ Noise or exception? — One person's garbage could be another person's treasure
- ❑ Methods: by product of clustering or regression analysis, ...
- ❑ **Applications:**
 - ❑ Fraud detection
 - ❑ Network intrusion detection



Evolution Analysis

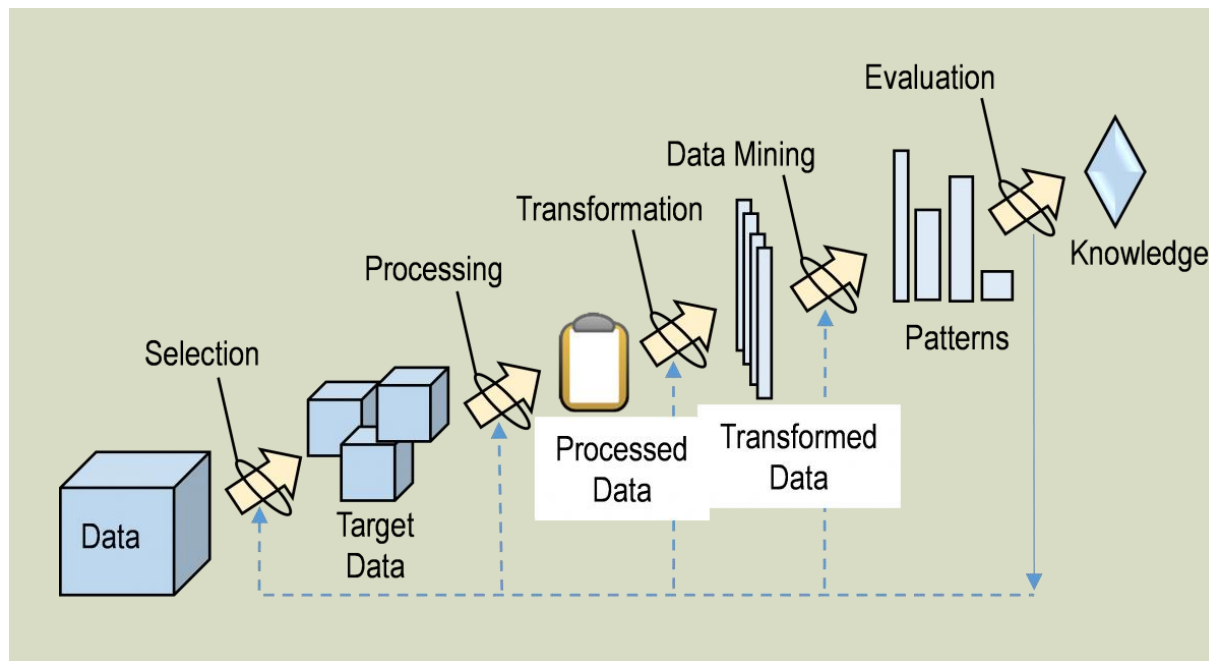
- ❑ Studies changes over time
- ❑ **Includes:**
 - ❑ Trend analysis
 - ❑ Sequential pattern mining
 - ❑ Periodicity analysis

Summary of Data Mining Functionalities

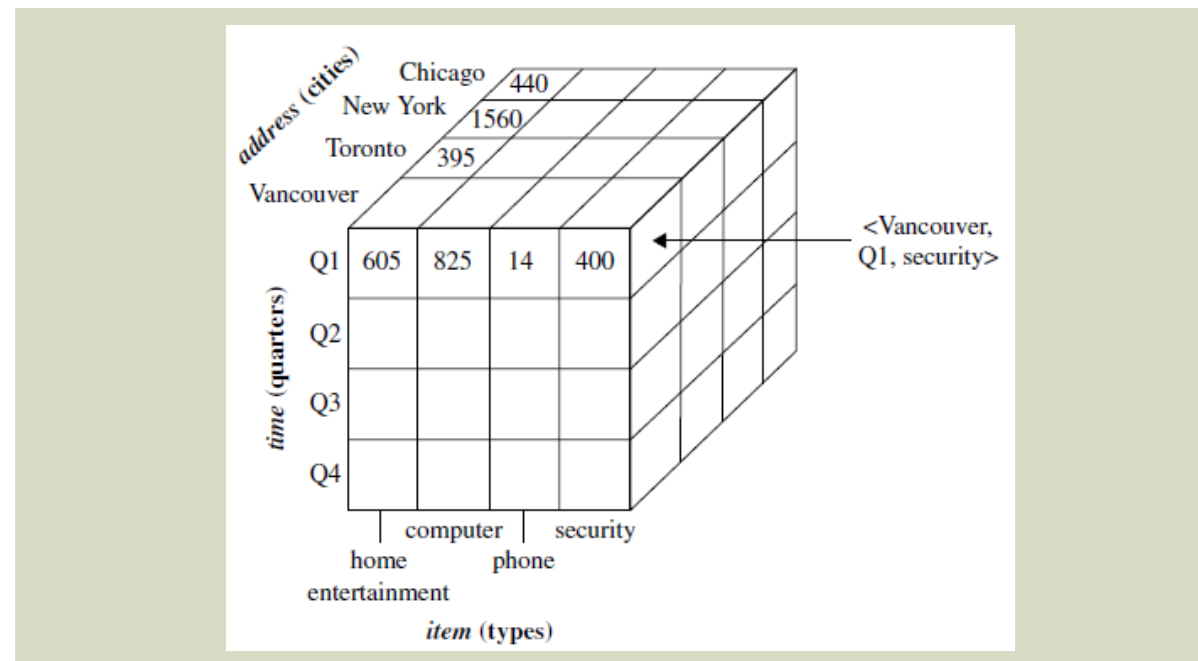
Functionality	Description	Learning Type
Characterization	Summarization	Descriptive
Association	Pattern relationships	Descriptive
Classification	Class labeling	Predictive
Prediction	Numeric estimation	Predictive
Clustering	Grouping	Descriptive
Outlier Detection	Anomaly detection	Descriptive
Evolution Analysis	Temporal patterns	Descriptive

Unit in a Minute

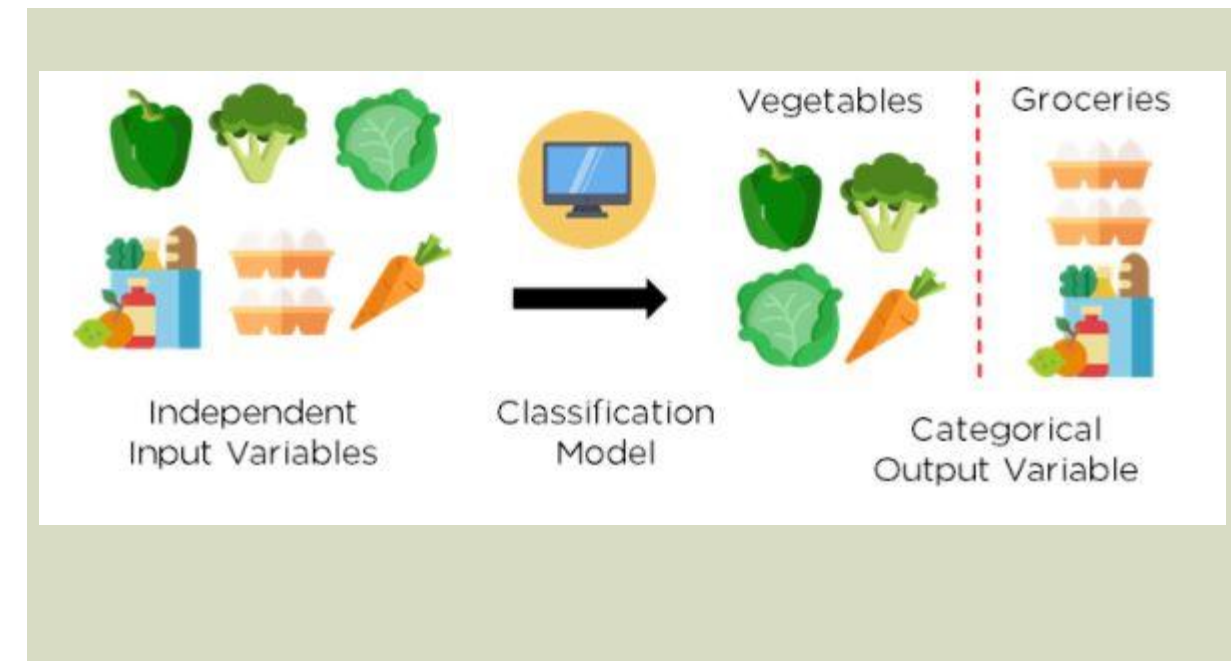
Knowledge Discovery



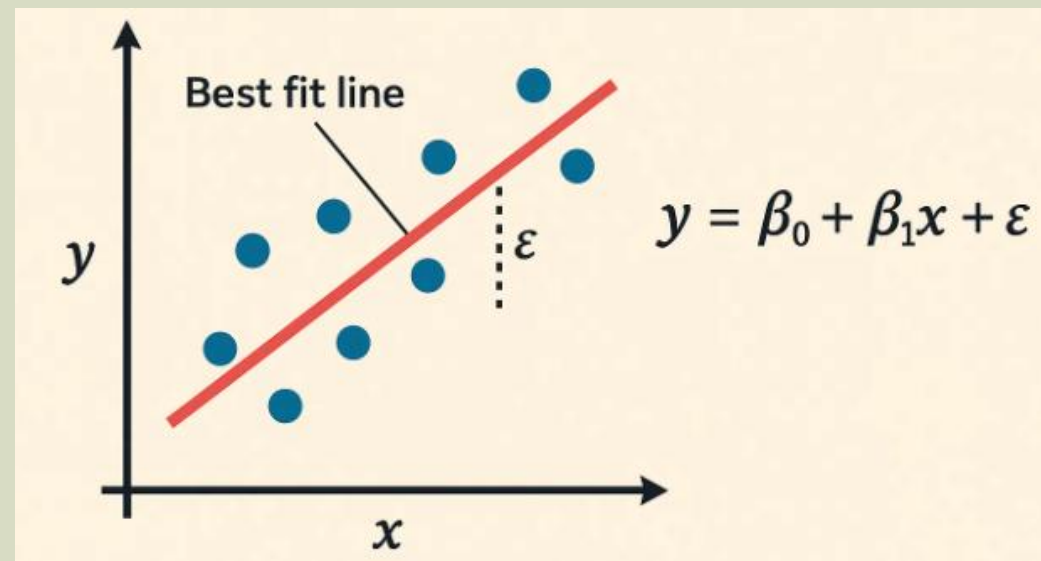
Data Warehouse



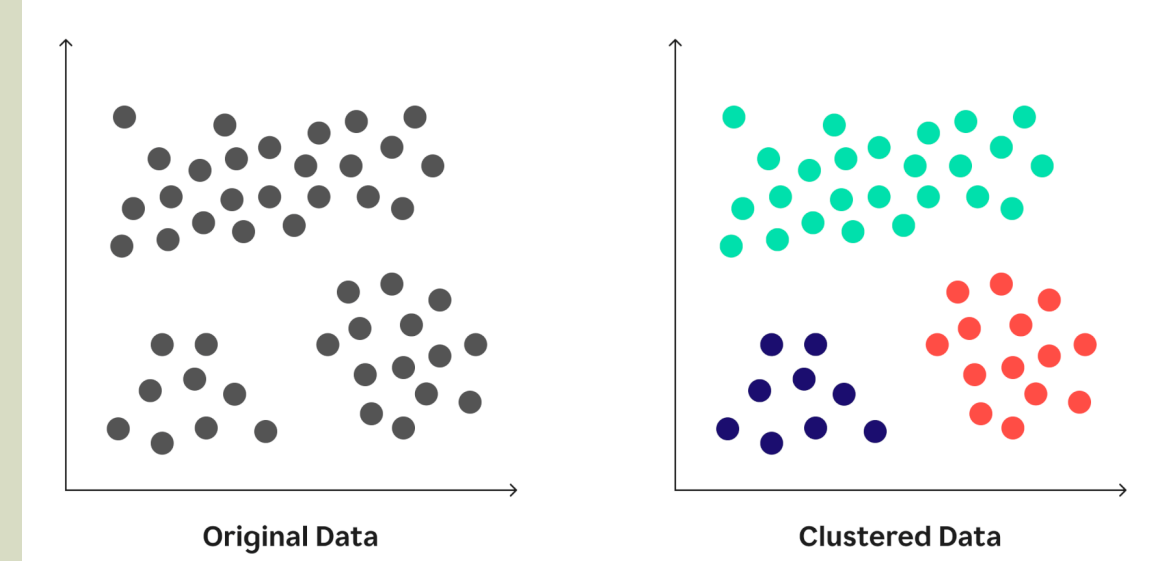
Classification



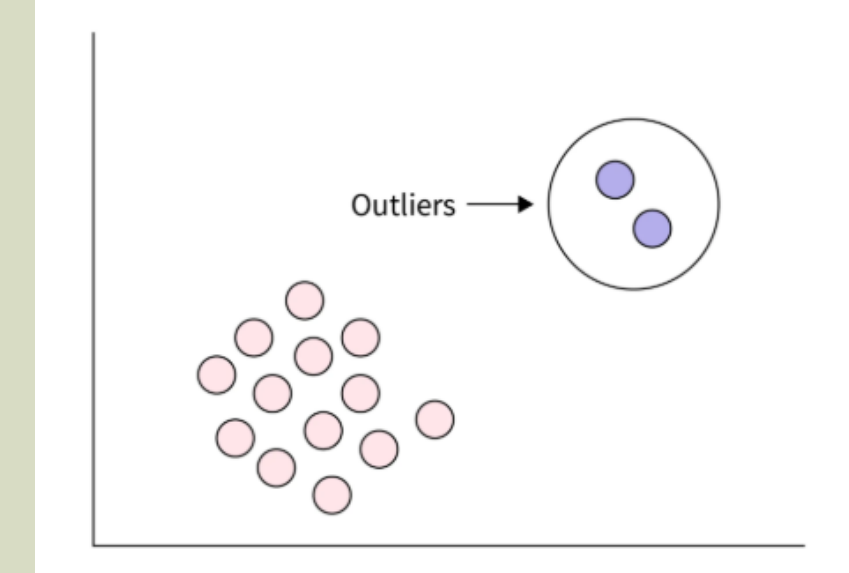
Regression



Clustering



Outlier Detection



References

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- Tan, P.-N., Steinbach, M., & Kumar, V. *Introduction to Data Mining*. Pearson.
- Author's lecture slides and instructional materials for CS 327 – Data Mining, NEMSU



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Thank you!

God Bless!

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